TRAN TRONG HIEU

BACKEND DEVELOPER

Thanh Tri, Hanoi, Vietnam | 0862 584 283 | tthieu.dev.02@gmail.com
tthieu22.github.io/portfolio

PROFESSIONAL SUMMARY

Backend Developer experienced in building enterprise backend systems, workflow-driven applications, and large-scale data processing solutions. Skilled in developing production-grade services, designing APIs, implementing complex business workflows, integrating multiple systems, and optimizing system performance, with a strong focus on scalability, reliability, and maintainable software.

PROFESSIONAL EXPERIENCE

LIFETEX TECHNOLOGY JSC — Backend Developer | Jul 2025 – Present

Enterprise Document Management System — SNP (Saigon Newport Corporation)

Tech Stack: NestJS, SQL Server, TypeORM, Camunda BPMN, Redis, WebSocket, MinIO, OAuth2, JWT, Java

- Developed NestJS backend services for an enterprise management system serving 7,000+ users, covering document management, meeting workflows, duty scheduling, reporting, and other internal business modules.
- Designed workflow-driven authorization using Camunda BPMN, RBAC, and ABAC to manage business processes, document lifecycles, and access permissions across organizational roles.
- Integrated NestJS services with Java-based file-processing modules, MinIO storage, and WebSocket notifications; supported large-scale DOCX, PDF, and Excel exports.
- Implemented OAuth2/JWT authentication and audit logging, and optimized SQL Server queries to improve security, traceability, and production performance.

Internal Data Synchronization Tool — SNP (Saigon Newport Corporation)

Tech Stack: Node.js, Worker Threads, SQL Server, SharePoint, Batch Processing, Parallel Processing

- Built a data synchronization and migration system to move approximately 10 TB of legacy data from SharePoint and multiple SQL sources to SQL Server.
- Processed more than 1 million documents and related tables containing 4–5 million records each, covering data from 2014 to 2026.
- Reduced synchronization time per document from approximately 6 seconds to 0.4 seconds using Worker Threads, data partitioning, and parallel processing.
- Shortened the overall migration timeline from nearly two months to 4–5 days; implemented retry strategies and transactional processing to improve fault tolerance and data consistency.

Digital Signature Service — SNP (Saigon Newport Corporation)

Tech Stack: Java, Spring Boot, USB Token, PDF Processing

- Contributed to upgrading an internally developed digital signature service, modernizing legacy code and adapting business logic to new requirements.
- Supported the DOC-to-PDF conversion workflow and USB Token-based digital signing for electronic documents.
- Helped embed signature images and digital certificate information into signed PDF documents.
- Supported integration of the upgraded signing service into the enterprise document management workflow.

LIFETEX TECHNOLOGY JSC — Backend Intern | Apr 2025 – Jul 2025

Document Management System — MPS (Ministry of Public Security)

Tech Stack: Node.js, Express.js, MongoDB, MongoDB Aggregation

- Built a backend service for sending and receiving real-time notifications in a document management system, synchronizing document updates and processing statuses across platforms, including mobile phones and computers.
- Developed APIs and business logic for notification workflows, helping users receive timely updates as documents progressed through processing stages.

- Developed reporting and statistical features using MongoDB Aggregation; maintained and extended APIs and resolved issues in a production system.

OVATHEME VIETNAM — PHP Developer | Jun 2024 – Mar 2025

- Developed and customized backend features for e-commerce, booking, and service websites.
- Implemented business logic and API integrations, worked with MySQL queries, and resolved issues found during testing and production maintenance.
- Maintained existing functionality and collaborated on feature updates based on project requirements.

PERSONAL PROJECTS

SmartCollab — AI Project Management Platform | Nov 2025 – May 2026

Tech Stack: NestJS, Spring Boot, Next.js, PostgreSQL, MongoDB, Prisma, RabbitMQ, Redis, Docker, Socket.io

- Built an AI-assisted project management platform using an API Gateway and backend services to support task management and real-time collaboration.
- Designed REST APIs and implemented JWT/OAuth2 authentication and role-based access control (RBAC).
- Used RabbitMQ for asynchronous processing and Redis/Socket.io for real-time features.
- Integrated AI-assisted requirement analysis and project planning, including WBS and task suggestions; containerized components with Docker.

GitHub: github.com/tthieu22/smart-collab | Demo: tthieu-smart-collab.vercel.app

Haniu — E-commerce Platform | May 2026 – Jun 2026

Tech Stack: Spring Boot, Next.js, PostgreSQL, Redis, Docker, Spring Security, JWT

- Built an e-commerce website using a Modular Monolith architecture, organizing product, category, cart, order, customer, and promotion modules.
- Developed REST APIs, JWT authentication and authorization, guest checkout, and order tracking.
- Applied Optimistic Locking to reduce inventory inconsistencies under concurrent updates.
- Used PostgreSQL JSONB and optimized JPA/database queries, with attention to N+1 query issues.

GitHub: github.com/tthieu-team/haniu | Demo: haniu.vercel.app

TECHNICAL SKILLS

Languages & Backend: Java, TypeScript, JavaScript, Node.js, NestJS, Express.js, Spring Boot, RESTful APIs, GraphQL
Databases: PostgreSQL, SQL Server, MongoDB, MySQL, Redis; schema design, query optimization, MongoDB Aggregation
Architecture & Processing: Microservices, Modular Monolith, API Gateway, Event-driven Architecture, Worker Threads, Batch and Parallel Processing
Workflow & Security: Camunda BPMN, RBAC, ABAC, OAuth2, JWT, audit logging
Real-time & Integration: WebSocket, Socket.io, RabbitMQ, MinIO, AI integration
Tools & Practices: Docker, Linux, Git, GitHub, GitLab, Postman, Prisma, Spring Data JPA, Agile/Scrum
Supporting Frontend: React, Next.js, TypeScript, Tailwind CSS, Zustand

EDUCATION

Thai Nguyen University of Information and Communication Technology | Sep 2020 – Sep 2025
Major: Information Technology — Software Engineering | GPA: 3.1/4.0
Graduation project: C# personal expense management application with AI-assisted note-taking.

ADDITIONAL INFORMATION

- Problem-solving and logical thinking; self-learning and technical research; system design and technical analysis; teamwork.
- English: Able to read and understand technical documentation and IT materials.
- Professional interests: Backend Technology Research, System Design, and AI Applications.
- Availability: One month after receiving an official offer.